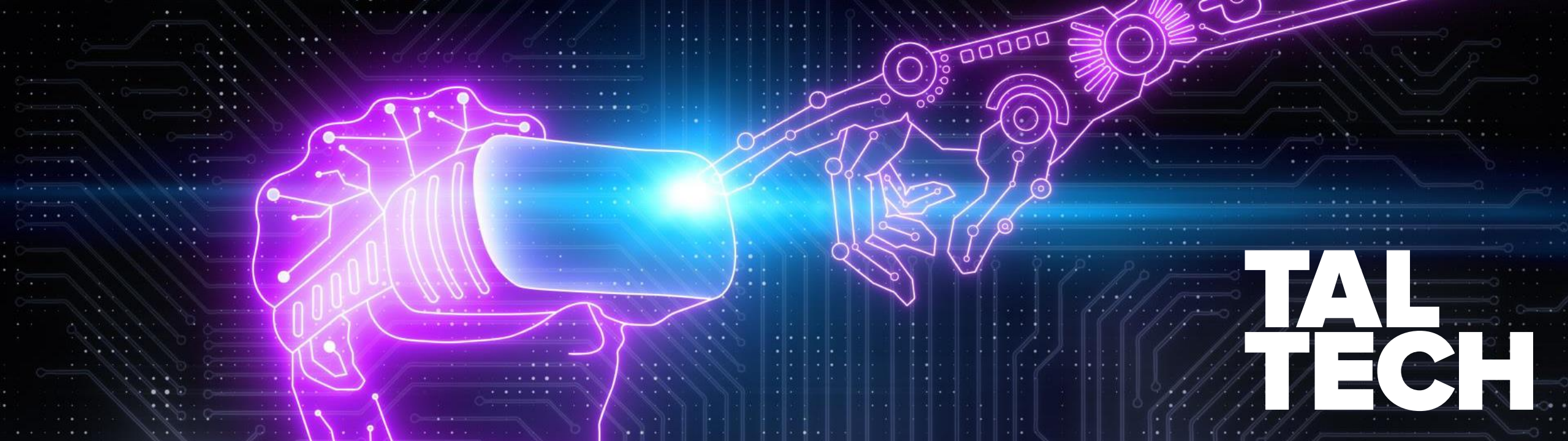




**TAL
TECH**

ITE4110: INFORMATION SOCIETY PRINCIPLES: TOWARDS E-GOVERNANCE

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E-GOVERNANCE TECHNOLOGIES AND SERVICES MASTER'S PROGRAM

QUICK COURSE ROUNDUP

- **Today's Agenda**

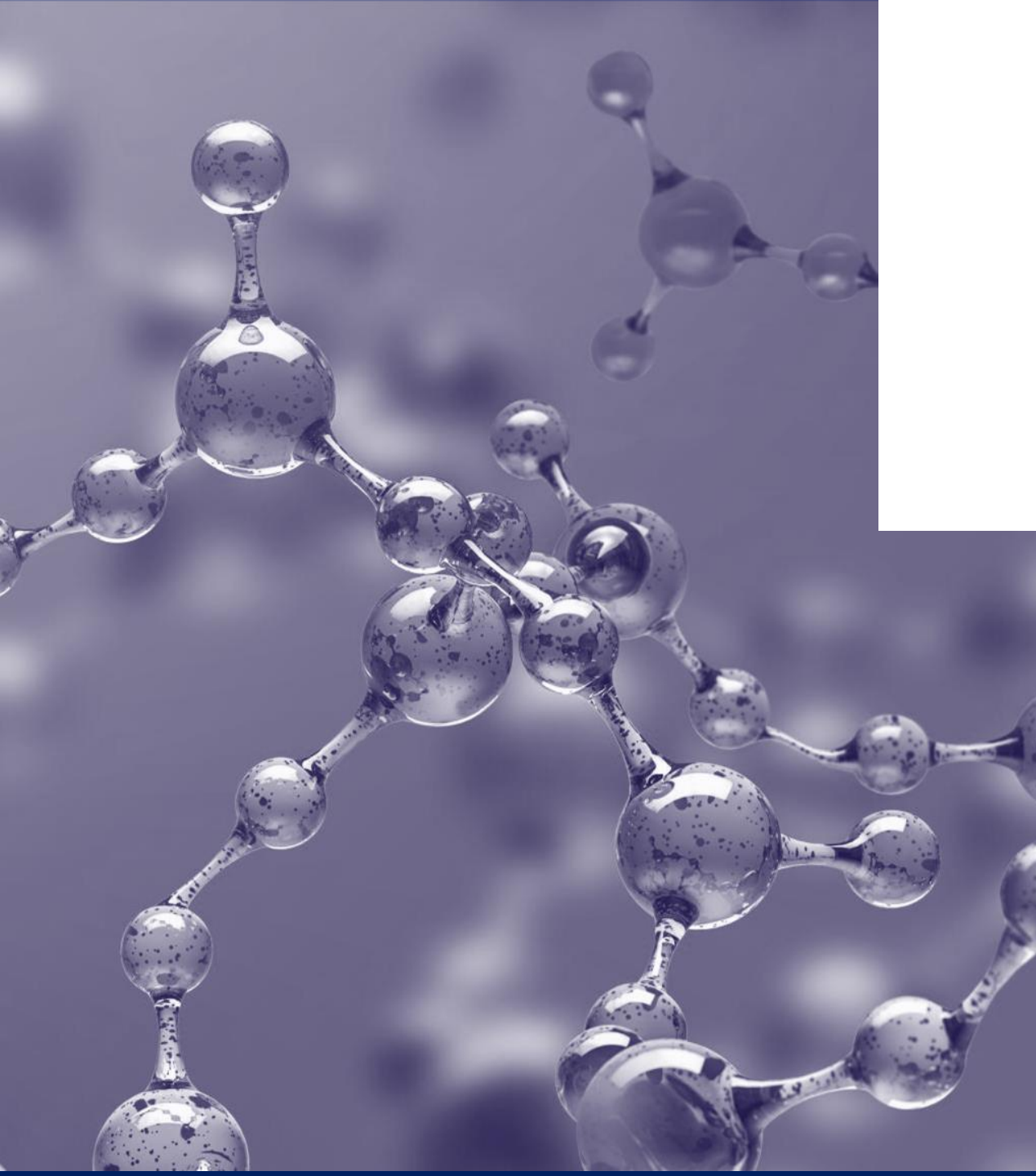
- Preliminary Case Study Overview
- E-Governance Sprint Week
- Change Management Primer – Eric
- Guest Lecturer: John O'Duinn

- **Task 2-3 Feedback**

- Will be provided by the end of the week

- **October 16th Class session**

- Data Governance/Open Data
- Guest Lecturer: Gledson Pompeu, Federal District, Brazil · Advisor on Innovation and Digital Government · Tribunal de Contas da União



STATE-OF-THE- ART PRELIMINARY CASE STUDY

EVALUATION CRITERIA

Introduction (8 points)

- **Purpose and Scope (4 points):**

The introduction clearly defines the purpose and scope of the case study. It provides a strong foundation for the reader to understand the context.

- **Engagement and Relevance (4 points):**

The introduction effectively engages the reader and establishes the relevance of the topic. It sets up the importance of the problem being studied.

2. Problem Statement (10 points)

- **Clarity and Definition (5 points):**

The problem statement is well-defined and clear, presenting the main issue(s) that the case study addresses.

- **Significance and Impact (5 points):**

The problem is significant and well-justified. The writer explains why the problem is important within the context of the field and what impact it has.

3. State of the Art (12 points)

- **Comprehensive Literature Review (6 points):**

The case study includes a thorough review of relevant literature, incorporating at least 10 academic sources. The literature review covers the key developments and current state of knowledge related to the problem.

- **Critical Analysis and Synthesis (6 points):**

The state of the art section critically analyzes the reviewed literature, identifying gaps, trends, and areas for further research. It synthesizes the information in a way that directly informs the case study.

EVALUATION CRITERIA

4. Discussion (10 points)

- **Depth of Analysis (5 points):**

The discussion section provides a detailed and insightful analysis of the problem. The analysis is supported by evidence from the state of the art.

- **Originality and Coherence (5 points):**

The discussion presents original insights or perspectives on the problem. The arguments are coherent and well-organized, contributing to the overall understanding of the issue.

5. Citations (5 points)

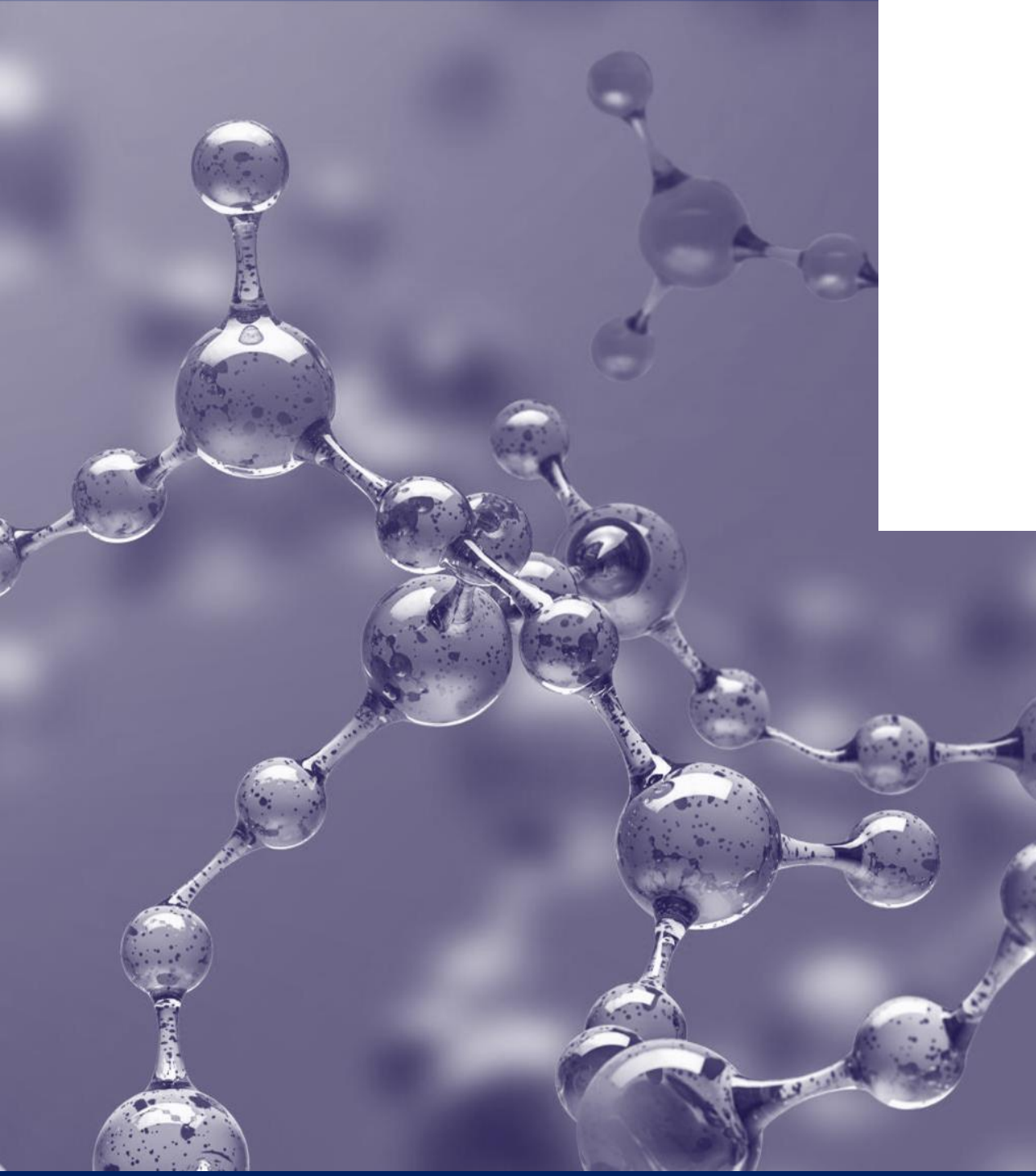
- **Accuracy and Formatting (3 points):**

All sources are cited (IEEE format) accurately in the text and listed in the bibliography according to the required citation style. Citations are complete and correctly formatted.

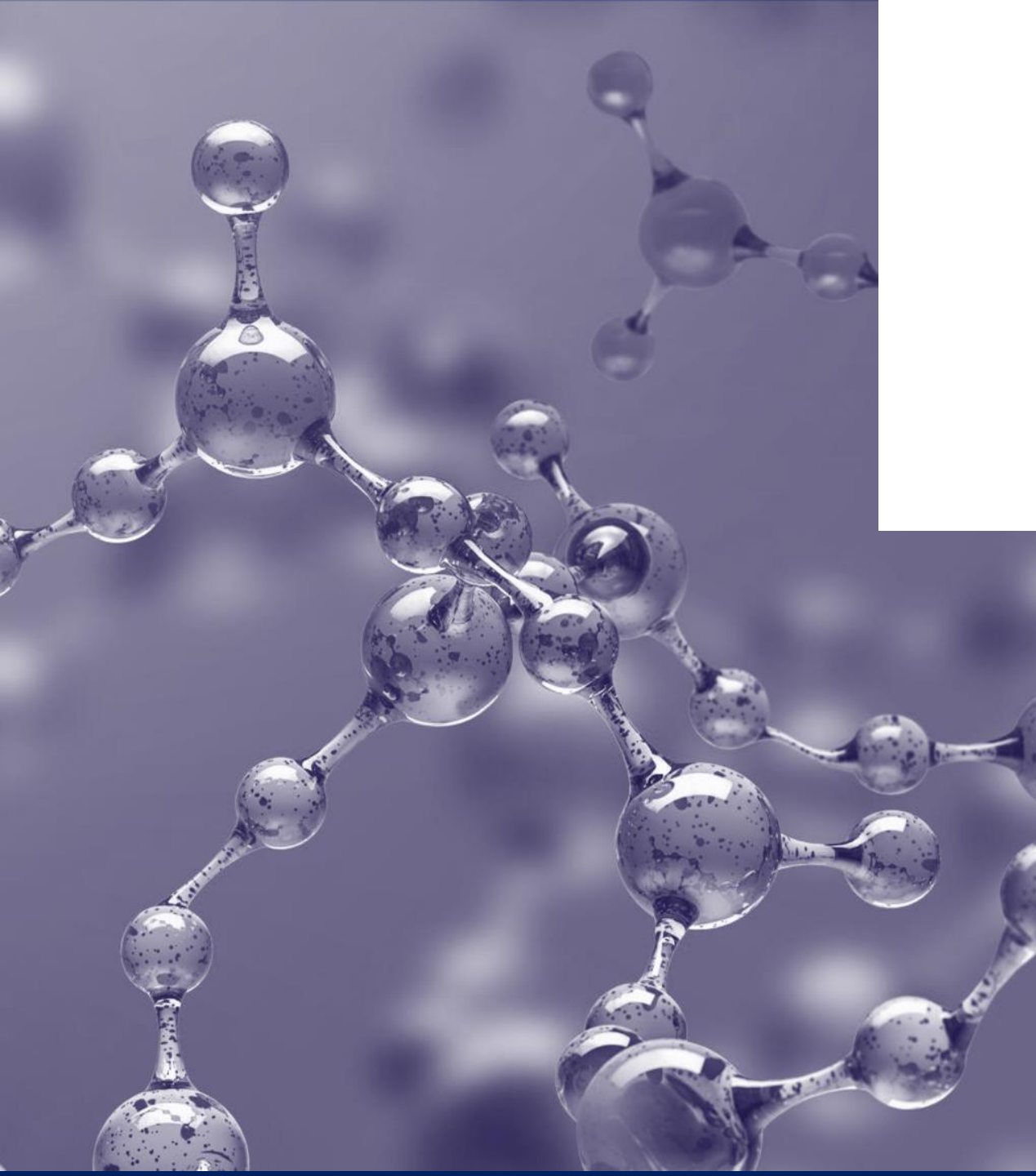
- **Integration and Relevance (2 points):**

Sources are effectively integrated into the case study, supporting the analysis and discussion. There is a balance between paraphrasing and quoting, with proper attribution given to all ideas.

TOTAL: 45 Points



STATE-OF-THE- ART PRELIMINARY CASE STUDY (EXAMPLE)



CASE SELECTION PROCESS

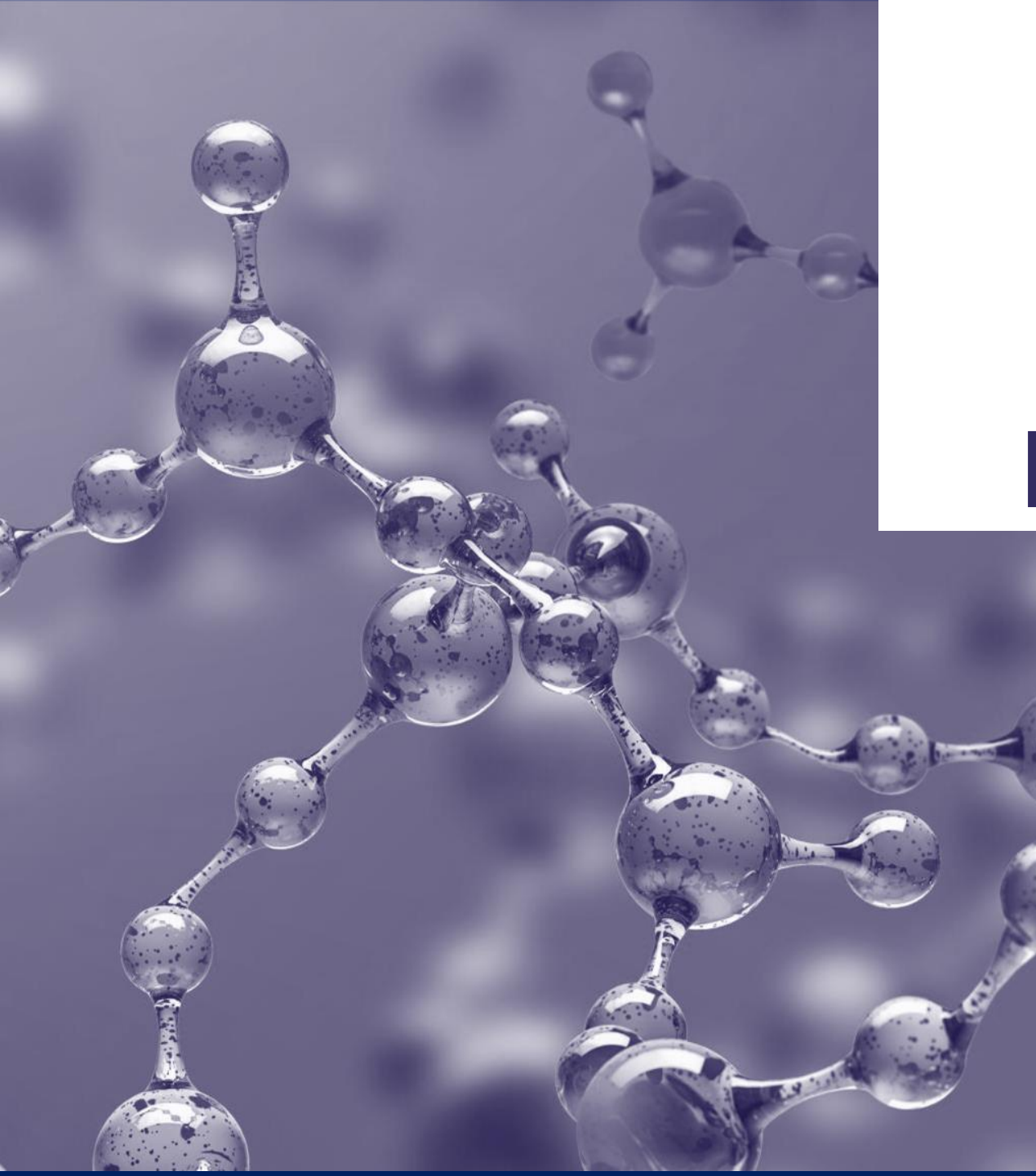
TWO PATHWAYS, LET'S DECIDE?

- We will register to the cases via Moodle, but we have two options:
- 1:
 - **Case Selection By Pillars and More General**
 - Pillar 1: AI in Public Sector Transformation
 - Pillar 2: Digital Identity & Trust Frameworks
 - Pillar 3: Cross-Border Interoperability & Standards
 - Pillar 4: Sovereignty, Resilience & Security
 - Pillar 5: Green & Inclusive Digital Governance
 - Pillar 6: Regulation & Policy Impact

TWO PATHWAYS, LET'S DECIDE?

- **PATHWAY 2:**
 - **More concrete and specific use-cases use register for individually**

Avoiding Vendor Lock-in and Legacy Lock-in: Ensuring Autonomy in the Era of Big Tech	This preliminary case study explores strategies for avoiding vendor lock-in and legacy lock-in within government digital services, focusing on the need for autonomy in the era of Big Tech. Case examples from Estonia, India, and the UK (or others) are analyzed to understand how governments mitigate dependence on proprietary software and cloud services. The study evaluates open-source solutions, interoperability standards, and procurement practices that help avoid being tied to a single vendor or outdated technologies. Key challenges should be examined, including balancing innovation with security and costs.
Compound Interest of Technical Debt: How to Manage This Dynamic?	This preliminary case study explores the concept of technical debt in the public sector and its long-term implications for digital governance. The study investigates how governments accumulate technical debt through outdated systems, short-term fixes, and failure to modernize legacy infrastructure. Case examples from the UK, Australia, and Germany are used to analyze strategies for managing and reducing technical debt, including re-platforming, adopting cloud services, and implementing agile development frameworks. The research examines the "compound interest" effect, where deferred maintenance increases future costs and reduces system agility.
Cost of Changing Regulations: Economic Impact and Digitalization	This preliminary case study analyzes the economic implications of regulatory changes, particularly in the context of digital transformation across the EU. The study evaluates how shifts in data governance, privacy, and digital infrastructure regulations affect businesses, governments, and consumers. Case examples from sectors such as financial services, healthcare, and tech illustrate the direct and indirect costs of adapting to new legal frameworks. Case examples from sectors such as financial services, healthcare, and tech illustrate the direct and indirect costs of adapting to new legal frameworks. The study also explores how outdated regulations can slow innovation and create technical debt.
Cross-border Interoperability between G2G in Developing Countries and Regions: ECOWAS, African Union,	This preliminary case study examines the dynamics of cross-border interoperability between government-to-government (G2G) systems in developing regions, focusing on organizations like ECOWAS and the African Union. The study evaluates the success of digital initiatives to improve cross-border collaboration in areas such as trade, security, and public health. Case examples from West Africa's cross-border data-sharing systems are analyzed to understand the challenges of achieving seamless interoperability, including regulatory fragmentation and infrastructural limitations. Interviews with regional policymakers and technologists provide insights into ongoing efforts to standardize data protocols and governance frameworks.



E-GOVERNANCE SPRINT WEEK!

05.11 – 08.11

WHAT'S THE PURPOSE?

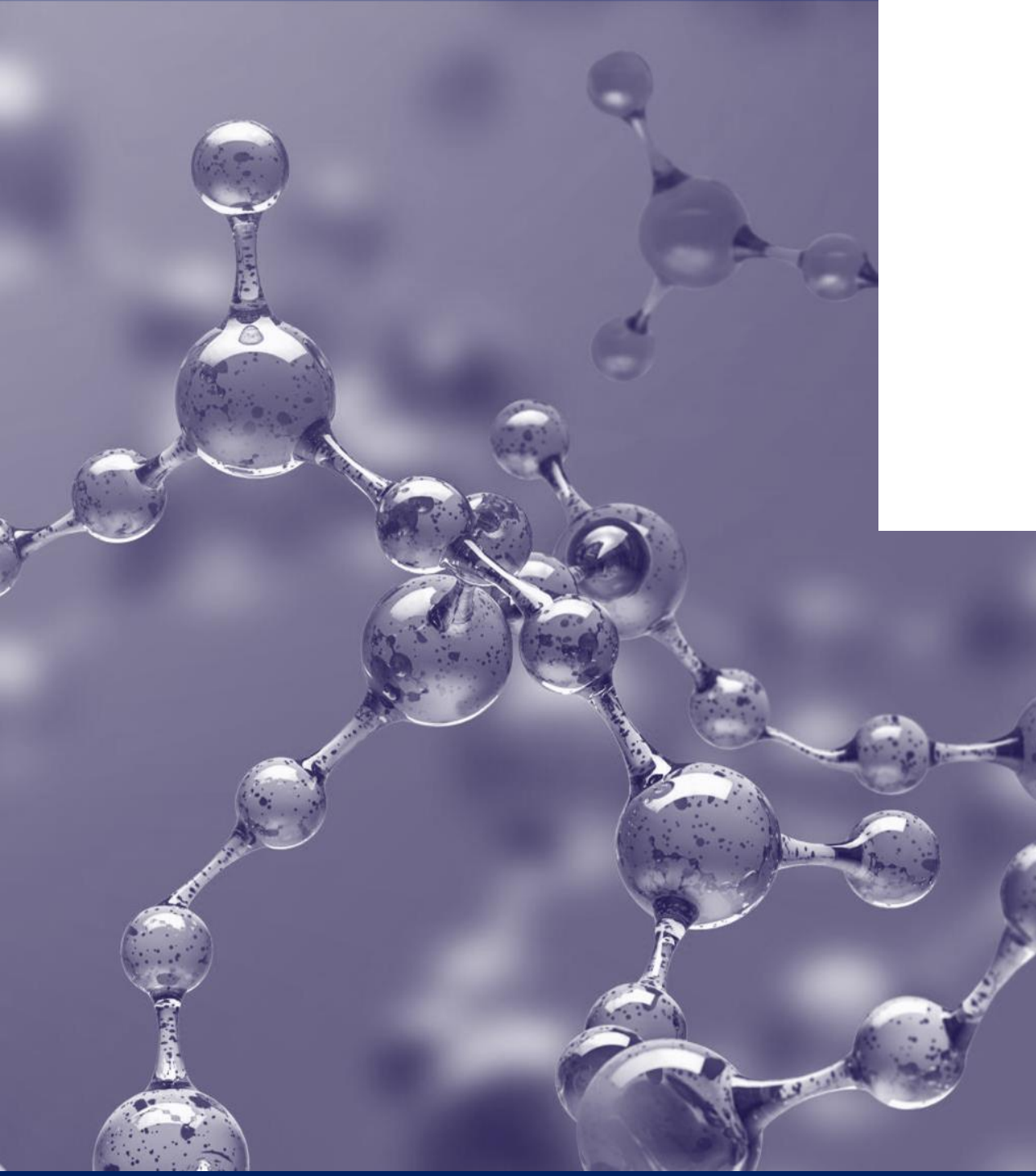
- Listen to practitioner perspectives on trending topics in the field
 - Discussion-oriented
- Built-in time to work on your preliminary case study
 - Guidance from the Next Gen Digital State team and your peers!
- Networking
 - Around 65 students from all over the world will be attending
 - Great opportunity to make connections!
- Output: 3 min presentation on your case:

SCHEDULE

Hands-on training week on spot @ Mektory			
Wednesday 05.Nov. 2025	Thursday 06.Nov.2025	Friday 07.Nov.2025	Saturday 08.Nov.2025
Room: SOC-211B	Room: SOC-11B	Room: SOC211-B	Room: SOC211-B
Welcoming Words Introduction to the Week (Eric Jackson, Silvia Lips, Mihkel Lauk) 10:00 - 10:45	The impact of Data Quality on Legal and Policymaking Rahul Sharma 10:00 - 11:30	Domain-Driven Design 10:00 - 11:30	Inspirational Talk/Peer to Peer Feedback Sessions 10:00 - 13:00
Preliminary Case Study Problem Solidification SWOT Analysis 10:45 - 12:00	Research Problem Development Tips for State of the Art 11:30 - 12:00	Tips for Discussion 11:30 - 12:00	
Lunch Break 12:00 - 13:00	Lunch Break 12:00 - 13:00	Lunch Break 12:00 - 13:00	
Trinidad Wiseman (Neuro UX) 13:00 - 14:30	Katrin Nyman Metcalf (AI+Ethics) 13:00 - 14:00	LLM's in the Public Sector 13:00 - 14:30	
Hands on Mentoring Session (Complete Swot Analysis) 14:30 - 15:45	Hands on Mentoring Session (Tips for the State of the Art) 14:00 - 15:00	Hands on Mentoring Session (Tips for the Discussion) 14:45 - 15:45	PARTY TIME/ NETWORKING 13:00 - 15:00
		AI Lightning Talk 15:45 - 16:15	
		Mentoring Session/Finalizing Presentations 16:15 - 17:00	

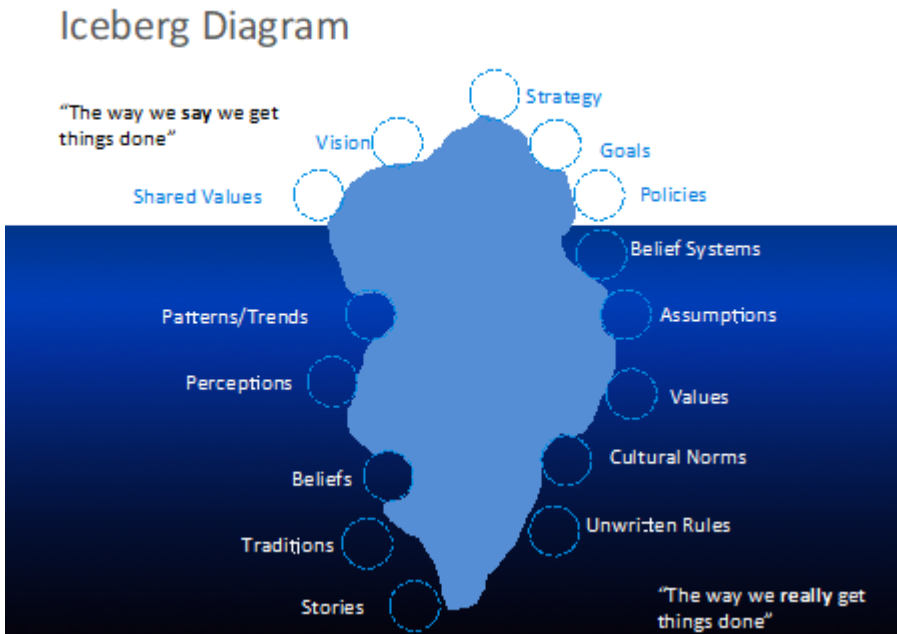
PRELIMINARY CASE STUDY SOLIDIFICATION SESSION OVERVIEW

- Nov 5th: How to execute a SWOT analysis of your case to give you a strategic viewpoint of the problem you are focusing on.
- **Learner Output:** Decide on case (finalization), and SWOT analysis of your case
- Nov. 6th: State of the Art: Tips for how to search for literature, identify keywords, identify quality literature sources, IEEE citation format
- **Learner Output:** at least 10 quality academic literature sources identified related to your case
- Nov. 7th: Tips for discussion, presentation finalization
- **Leaner Output:** writing time/2-3 minute mini presentation preparation for Saturday
- Nov 8th: Inspirational Speaker and presentation
- **Leaner Output:** 2-3 minute mini presentation of your case to your peers/NGDS team as we will divide into groups

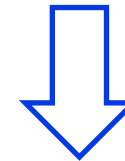
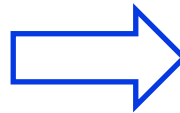


CHANGE MANAGEMENT PRIMER

Governments need a whole of society approach to change



Change Management should focus on the
public administration reform.



through a whole of society approach

Government as a
spider net of
dependencies

Rising challenges in digital societies are real issues we must face head-on

POLITICAL CLIMATE

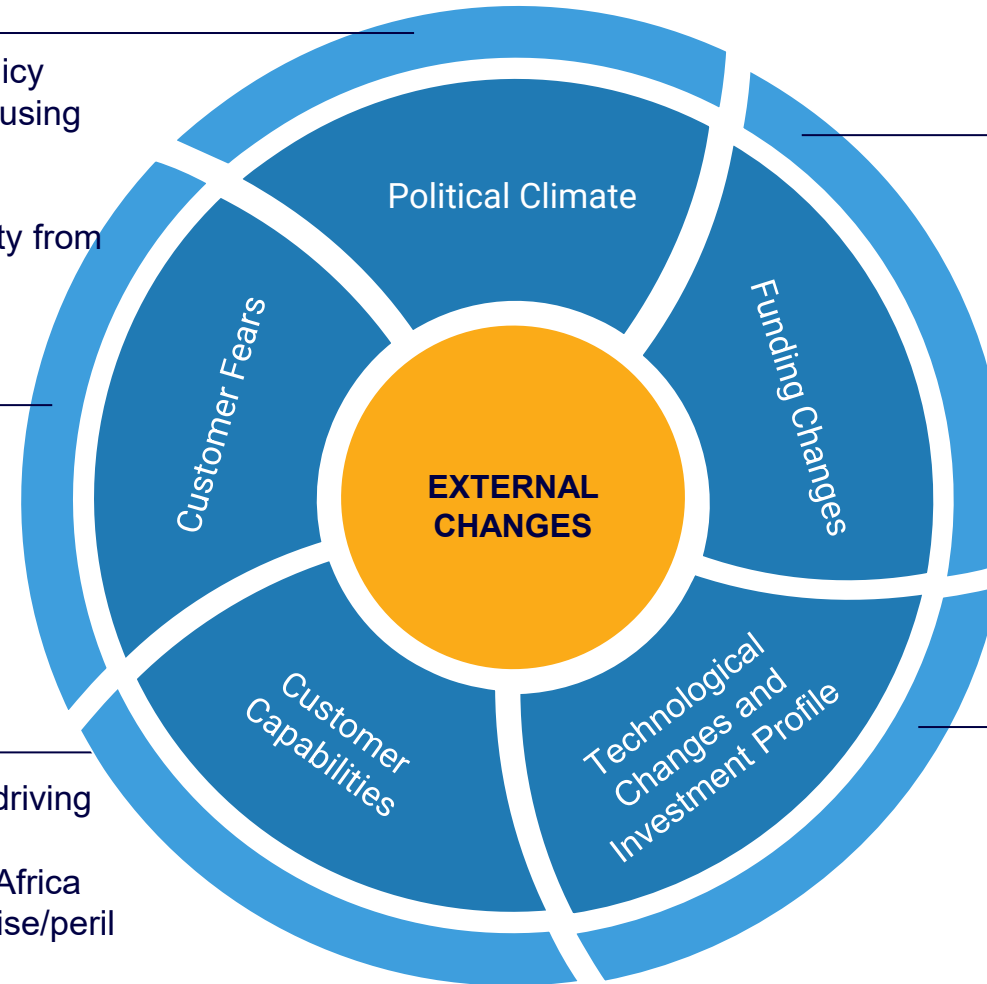
- Rising nationalism being linked to tech policy
- Country driven economic transformations using digital
- Increase in “big aid” skepticism
- Pressure to address rising citizen insecurity from climate, food, health – linked for first time

CITIZENS FEARS

- Increased social isolation and echo chambers
- Increasing dependence on technology to do basic tasks affecting jobs
- Increased spread of rumors, data misuse fueling societal discord

CITIZEN MISMATCH

- Growing generation of digital first citizens driving new challenges
- Growing youth populations, particularly in Africa
- New data sources coming online but promise/peril not yet well understood
- Lack of comfort with technology generally in public sector; struggling to adapt to technology changes



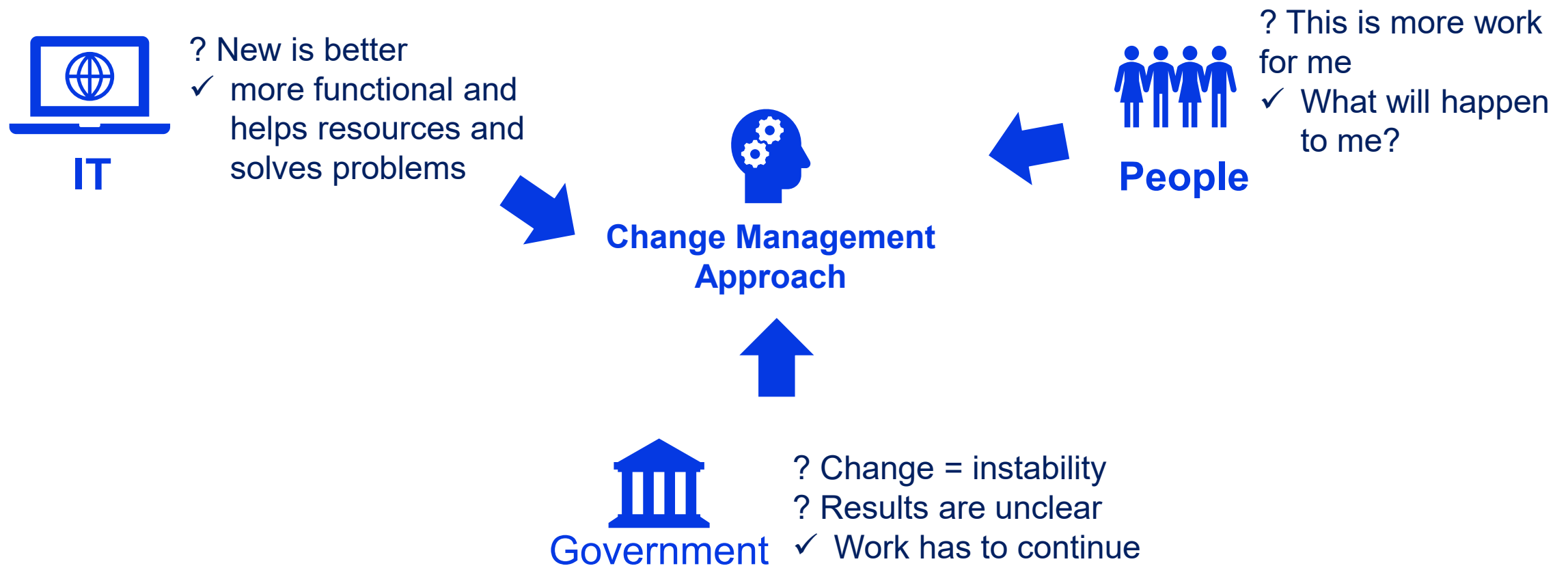
FUNDING CHANGES

- Decreased size of AID funding in total and by award;
- Increased contribution of countries and private sector to achieve development goals
- Countries looking to control and possibly monetize their data
- Promise and perils of zero-rated services and internet infrastructure
- Tax policy

TECHNOLOGICAL CHANGES AND INVESTMENT PROFILE

- Cloud hosting, AI and ML, 4-5 G, blockchain
- Open source becoming mainstream for Big Tech
- Recognition by companies that Africa/LatAm/Asia are key to profitability

For a successful implementation the different perspectives on change (management) have to be considered



Why Governments need a Change Management approach?

A Change Management Approach establishes how changes will be **proposed, analyzed, accepted/rejected, implemented, monitored, controlled, and documented.**

Change Management (CM) aims to accomplish several things including:

- ❖ Encourage and guide the government in how to effect sustainable change;
- ❖ Facilitate discussions on problem identification and possible scenarios for solution;
- ❖ Introduce innovative concepts and ways in which effective change works;
- ❖ Help develop and highlight steps the government needs to take to make change happen;
- ❖ Help governments analyze and break down complex problems into manageable components that can be solved more effectively;



Whole-of-society approach to managing change

In managing the change process, some of the areas that should be focused on throughout the digital transformation process are:

- **the problems** (the right problem is being solved)
- **benefits/values of the change, requirements for change (personnel/financial/political resources),**
- **solutions** (the right solutions to these problems are being developed – based on requirements),
- **impact/changes** to policies/processes
- **alignment** with government ICT development/digital agenda goals,
- ensuring that the **change/transition process** is undertaken in an effective, efficient, and controlled manner.

Cultural difference between IT and Government

IT



“yes” / “no”

result-orientation

stress on solutions

vs

Government



“maybe” / “it depends”

process-orientation

stress on procedures

Several wrong assumptions about change management are existing

Assumption

- Only **leadership support** is sufficient
- **Fundamentals exists** if there is an agreement to do something digital
- Digital transformation is a **job of the IT department only**



Reality

- Leadership support is **necessary but not sufficient**
- Fundamentals like internet access, digitalization of data, interoperability and digital identity **have to be addressed first**
- IT department = technical work but **other departments are needed**: financing for funding, HR for capacity building, PR for awareness, process owner for real-life situations, legal for regulatory changes

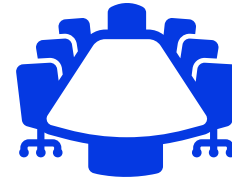
Some common assumptions on the other hand correspond to reality



IT can help to make the government become better



It is possible not only to change the IT system but also bureaucratic culture



Main challenge is to change the attitude of the middle management

If the government and IT cooperate the digital transformation will be successful



- Managing the change is possible but not easy
- Cooperation between government and IT is necessary to better understand and imbedded inherent limitations and dangers. IT needs support to know and understand way of the government
- Government does not need to do specific technology steps, there is plenty of work for bureaucrats even before



DISCUSSION CHECKPOINT

Theory of Change assumptions

- **There are multiple possible pathways to digital transformation, including but not limited to beginning with undertaking a whole-of-society approach**
- Strong civil society involvement is necessary for digitation transformation to ensure that governments do not misuse citizen data at some future point
- Global and national leaders currently do not invest in digital transformation agendas in part because they do not understand what digital and data solutions exist to support this and do not know how to begin the process
- More country exemplars will drive new countries to undertake digital transformation
- Evidence alone is not enough to drive behavior change; it must be accompanied by coaching and political will
- Timing and conditions are right for development actors to coordinate efforts and money.
- National-level building blocks can be deployed at scale and linked using standards to existing sectoral investments
- There is a market failure in developing interoperable, scalable digital global goods, partly due to competition for project-specific donor funding
- If demand is aggregated for the social sector and a reasonable profit margin is assured, that aggregate volume will be sufficient to spur new digital and data investments from tech providers in low low-income markets.

DISCUSSION QUESTION

- **Have you ever managed or participated in change management processes at you work?**
 - **How did this experience go?**
- 

The Problem Driven Iterative and Adaptive (PDIA) approach to Change Management

- PDIA is very useful in projects where problems are complex and have many unknown variables such as resources required, solutions that will work, or other underlying factors.
- As government projects often fall into this category, PDIA is a logical approach for building administrative capability, not only by helping stakeholders use PDIA to implement ICT solutions, but also by developing their ability to manage change in the future.



Identify measurable problems that motivate stakeholders to drive change and find effective solutions collaboratively.

Problem Driven



Employ continuous evaluation to refine solutions based on feedback, enabling informed adaptations to the change strategy.

Iterative Learning



Create tailored solutions that respect local contexts, ensuring interventions align with specific cultural and operational needs.

Contextual Adaptation

Benefits of PDIA approach

PDIA focuses on **response rather than critique** on building up the functionality and capability of an organization to deliver on complex problems by digging deep into why a problem exists.

Using PDIA allows a team to focus on solving problems as they emerge by:

- breaking them down in manageable parts
- determine the type of capabilities needed for successful change
- analyze the activities that need to happen in order to solve a problem
- ascertain the resources required
- identify and shorten implementation gaps by looking at the local context of what is required to get from point A to point B.



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Objectives of Change Management



Guide government efforts to implement changes that are lasting and beneficial for society.

**Encouraging
Sustainable
Change**



Engage GoSL in dialogue to identify challenges and explore potential solutions collaboratively.

**Facilitating
Problem
Discussions**



Introduce effective methodologies that promote successful and sustainable change practices.

**Introducing
Innovative
Concepts**



Enhance GoSL's competence in managing change and executing strategies effectively.

**Developing
Administrative
Capability**